



Python Programming Foundations

Week 8 - Capstone Build, Justification, & Presentation
Day 1: "Build Expectations & Validation Evidence"

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Finishing Code Is Not Enough



A capstone is not finished just because the code runs once.



It should run inside approved scope, produce meaningful behavior, and include evidence that important behavior was checked.

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Code runs once

```
python process_data.py
Processing data...
Calculating results...
Done
Total records: 1,248
Total amount: 32,945.18
```



Code runs with evidence

```
python process_data.py --evidence
Processing data...
Calculating results...
Done
Total records: 1,248
Total amount: 32,945.18
Evidence saved: runs/2025-05-20_10-10-30/
```



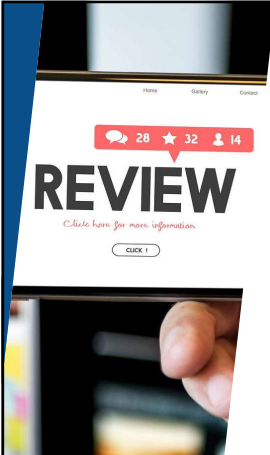
Validation
Evidence package created and verified.
Results can be reproduced and reviewed.

Finishing Code Is Not Enough

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From Approved Scope To Build Evidence



REVIEW
Click here for more information
CLICK 1

- ▶ Week 7 approved the project direction.
- ▶ Week 8 asks: what did you build, how do you know it works, and how can someone else understand how to run it?

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Approved Scope → Build Evidence



1. APPROVED SCOPE

Clear goals and boundaries for this project.

- Problem and purpose
- Core features to build
- What is in scope
- What is out of scope
- Success measures



2. WORKING BEHAVIOR

Smallest useful build that shows the project works.

- Core behavior works
- Input produces output
- Key logs is working
- Runs without errors
- Matches the scope



3. EVIDENCE AND RUN GUIDANCE

Clear proof and simple steps for anyone to run it.

- Files and organization
- How to run the project
- Sample input and output
- What to check
- Notes on results

Scope is approved. Behavior works. Evidence is clear.
Project is ready.

From Approved Scope To Build Evidence

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What Counts as Success Today

- Stay inside approved scope
- Identify the most important behavior
- Validate that behavior with real evidence
- Preserve run instructions
- Record meaningful AI help if used
- Prepare to explain the work in Assignment 16

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Evidence Can Be Small And Real

Validation evidence can be simple:

- ▶ expected versus actual output
- ▶ one successful run
- ▶ one boundary or error case
- ▶ short validation note
- ▶ screenshot or copied console output if required

Small evidence is acceptable when it is real.

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Evidence Can Be Small And Real

Validation in Three Simple Ways

- 1. EXPECTED VS ACTUAL**
Compare what should happen with what actually happened.
Input: 5
Calculation: $(5 * 4) + 7$
Expected: 27
Actual: 27
Match: ✓
- 2. SUCCESSFUL RUN**
The program runs from start to finish and produces results.
Python example output:
Enter name: Alex
Hello, Alex!
Total items: 3
Average score: 88.5
Program finished successfully.
- 3. BOUNDARY CHECK**
Check the edges to make sure the program handles limits.
Test: Highest allowed value
Input: 100
Output: Accepted
Test: One over the limit
Input: 101
Output: Rejected

Small evidence is still evidence.

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Run Instructions Are Part Of The Project

A project should tell someone how to run it.

Good run instructions include the file to run, required data files, setup notes if needed, and the expected result.

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Run Instructions Are Part Of The Project

A Clear Project Folder

Organized files + a simple README help anyone run and understand your project.

PROJECT FOLDER

- main.py: Main program file
- data.csv: Sample data file
- utils.py: Helper functions
- README.md: Run instructions and info
- output_example.txt: Example output

README.md

- 1. FILE TO RUN**
Run this file:
python main.py
- 2. DATA NEEDED**
Place this file in the project folder:
data.csv
- 3. EXPECTED RESULT**
You should see output similar to:
output_example.txt

Clear files. Clear steps. Clear results.

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AI Notes Belong With The Build



If AI helped during the build, record it while the work is fresh.



Useful notes include what AI suggested, what you accepted, what you changed, and how you verified the result.

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AI-Use Note: Student Ownership

AI can support your work. You stay in charge.

1 AI SUGGESTED What did the AI suggest or provide?	2 I ACCEPTED OR CHANGED What did you accept, change, or combine?	3 I VERIFIED How did you check that it works and is correct?	4 I CAN EXPLAIN Can you explain this in your own words?
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Your work. Your understanding. Your responsibility.
Use AI wisely. Show your thinking. Own your results.



AI Notes Belong With The Build

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Demo

"Capstone Validation & Run"

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What to Watch For: Validation Evidence

- ▶ Watch the behavior first.
- ▶ Then watch the evidence that supports the claim that the behavior works.

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From Behavior to Evidence

A small program run becomes meaningful when you capture and check the results.

BEHAVIOR RUNS The program does what it is supposed to do. Program: total_calculator.py Run Output Enter first number: 12 Enter second number: 8 Enter third number: 5 Total: 25 Checkmark The program accepts input, does the calculation, and shows the result.	EVIDENCE NOTE Capture what happened and check that it matches. EXPECTED RESULT 12 + 8 + 5 = 25 ACTUAL RESULT Total: 25 MATCH The actual result matches the expected result. Checkmark This run shows the program works as intended.
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Small behavior. Clear evidence. Strong project.

Demo: Validation Evidence

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Demo: Run Instructions

Run instructions should reduce confusion.

They should tell someone:

- ▶ what to open
- ▶ what to run
- ▶ what files are needed
- ▶ what successful output looks like.

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Run Instructions (README)

Quick steps to run this project and see it work.

README – Run Instructions

- 1. OPEN THIS FILE**
Open the main program file in this project.
`main.py`
This is the file to start.
- 2. RUN THIS COMMAND**
Run this command in a terminal from the project folder.
`> python main.py`
- 3. EXPECTED OUTPUT**
You should see output similar to this.
Welcome to the Program!
Total: 25
Thank you!

Tip: If something looks different, check your input or data file.

Demo: Run Instructions

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Assignment #15 Build Checklist

Use this checklist to make sure your final build work is complete and ready.

A15 Build Checklist

- 1. APPROVED SCOPE**
My work stays inside the approved project scope.
- 2. WORKING CORE**
The core behavior works as intended with real data or input.
- 3. ORGANIZED CODE**
My code is clean, named well, and easy to follow.
- 4. VALIDATION EVIDENCE**
I have evidence that shows expected results and actual results.
- 5. RUN INSTRUCTIONS**
Someone else can open and run my program successfully.
- 6. AI-USE NOTES**
If I used AI help, I recorded what I used and how I used it.

★ Complete work. Clear evidence. Confident explanation.

- ▶ approved scope
- ▶ working core behavior
- ▶ organized code
- ▶ data files if needed
- ▶ validation evidence
- ▶ run instructions
- ▶ AI-use notes if used

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Today's Work Target

Before class ends, try to have:

- > One important behavior working or clearly attempted
- > One validation note started
- > README/run instructions started
- > One next-step decision identified

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Day One Work Target

Make steady progress. Build a strong start.

- 1. WORKING OR ATTEMPTED BEHAVIOR**
Get the core behavior working or show your best attempt.
- 2. VALIDATION NOTE STARTED**
Begin a note that captures expected and actual results.
- 3. README/RUN INSTRUCTIONS STARTED**
Start a short guide so someone else can open and run your program.
- 4. NEXT DECISION IDENTIFIED**
Choose the next small decision that moves your project forward.

Small steps today lead to strong results tomorrow.

Today's Work Target

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Closing Success Check

By the end of today, you should be able to say:

"I know the most important behavior in my project, and I have started proving that it works."

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Success Check

Focus on the core. Show the evidence.

Project Core Behavior

My program does the main thing it is supposed to do.

Example: Daily Total
The program adds the item prices and shows the total.

Apples	2.50
Bread	1.75
Milk	3.25
Total:	7.50

Evidence Started

I have begun capturing evidence that it works.

Expected Result	Total should be 7.50
Actual Result	Total is 7.50

Notes
Tested with sample prices. Result matches expected.

★ I have started proving that it works.

Success Check

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